

## 52nd CIRP Conference on Manufacturing Systems

## Overview of the State of the Art in the Production Process of Automotive Wire Harnesses, Current Research and Future Trends

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Today's automotive wire harness manufacturing process involves a high volume of manual work. However, current and future application requirements like miniaturization of electronic components, monitoring of process parameters, increasing demand of process documentation and rising wages call for a higher degree of automation.

This paper presents the state of the art in wire harness assembly, reviews currently available special purpose machinery and equipment as well as their limitations and highlights the challenges of automation. It discusses potentials for automation, current innovations and research and concludes with an overview of current trends in the automotive industry.

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Peer-review under responsibility of the scientific committee of the 52nd CIRP Conference on Manufacturing Systems.

**Keywords:** automotive engineering; vehicle electrical system; assembly systems; manufacturing automation; design for manufacture

**1. Introduction**

A wire harness is an assembly of electrical cables which is used in automobiles to connect electronic components, control units, sensors and actuators. The single wires which transmit signals and electrical power are bundled by straps, cable ties, cable ducts, adhesive tape and the like. The cable ends are fitted with electrical terminals, optionally equipped with wire seals and inserted in electrical connectors, plugs (male-ended) and jacks (female-ended). Connector housings with locking mechanism and seals are used to protect the integrity of the electrical connections against environmental influences such as shock and vibration, moisture, water spray or dust (see Fig. 1). [1–3]

Automotive wire harnesses are generally supplier products which for European car manufacturers are manufactured in countries like Romania, Poland, Tunisia and Morocco [4, 5]. There are several independent wire harnesses in an automobile.

These include interior electrics, engine control system, dashboard, doors and seats, among others [1]. Since the invention of automobiles the number of electrical components and, thus, the number of wires and total weight has steadily risen. This trend is due to the increase in functionality [2]. Today's vehicles usually have several thousands of single wires with an aggregated cable length of several kilometers [6]. The total wire harnesses weight of a 2018 Audi A8 adds up to 36 kg [7].

Compared to all other wire harnesses in an automobile, the engine wire harness places special requirements in regard to higher temperature, vibration, media and particle resistance compared to remaining wire harnesses in an automobile [2]. To illustrate production processes, we use an engine wire harness as example: A variant of the Mercedes-Benz OM654 engine wire harness consists of 54 plugs, 116 single wires, has, lying flat, a dimension of 4 m<sup>2</sup> and weighs 1.5 kg.

Today's high degree of customizability in vehicles results in

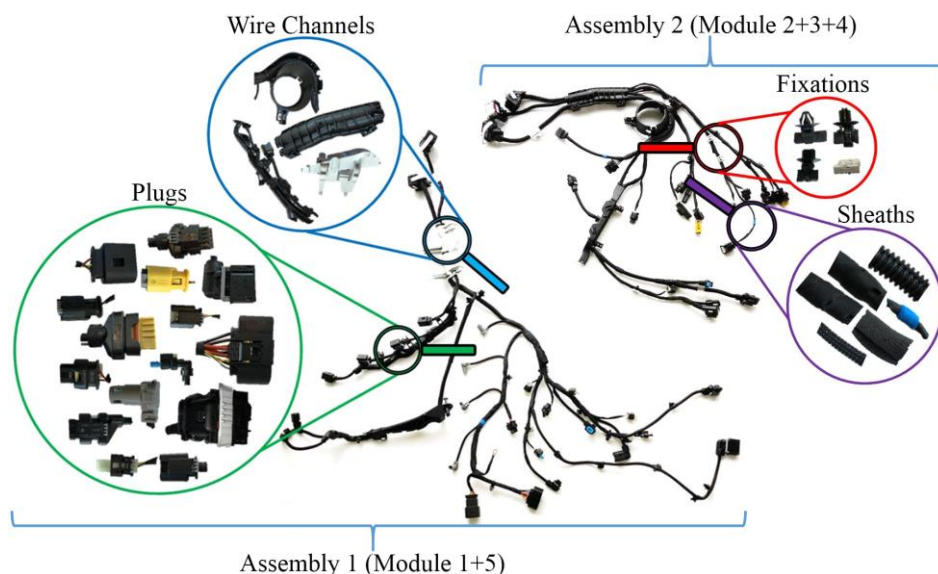


Fig. 1. Mercedes-Benz OM654 Engine Wire Harness and Variant Diversity of Selected Components.

a large diversity of wire harnesses in small batch sizes (down to 1).

A wire harness is usually designed in accordance to electrical, geometric and environmental requirements. The assembly of a wire harness—the main step in the production process—is carried out on a special assembly board which is used to form the wire harness to geometric specifications. In order to simplify the assembly process, a wire harness is often broken up into modules, which can be pre-assembled individually. The final assembly of the modules as well as the completion of the wire harness is again carried out on an assembly board [2, 3].

In spite of advances in production technologies and automation, in general, today's automotive wire harness manufacturing processes are still characterized by a high degree of manual work. Due to the variety of different work steps involved, only parts of the assembly (e.g. certain pre-assembly steps) have been automated using special machinery and equipment. However, since the remaining assembly steps can, however, be carried out by low-skilled workers, manual manufacturing in low-wage countries remains more cost-effective than automation. [6, 8]

Current and future application requirements like autonomous driving technologies, miniaturization of electrical terminals, monitoring of process parameters, increasing demand of process documentation, and a permanently high cost pressure – the wire harness is one of the most expensive produced parts – call for a higher degree of automation [9–12].

This paper reviews the state of the art in wire harness assembly. It is organized as follows: Sec. 2 describes the exemplary wire harness used in this work. Sec. 3 points out typical design requirements for automotive wire harnesses. Sec. 4 describes wire harness production processes. Sec. 5 reviews currently available automation machinery and equipment as well as their limitations and potentials. Sec. 6 highlights automation challenges as well as their requirements. Sec. 7 discusses innovations and current research. Sec. 8 provides an outlook as well as necessary research. Sec. 9 lists future requirements and automation supports. Sec. 10 provides

a short summary of the current state of the art and concludes with an overview of current trends.

## 2. Exemplary Wire Harness

In this paper the wire harness of the OM654 engine, a diesel engine from Mercedes-Benz introduced in 2016, serves as illustrative example. Fig. 1 depicts this engine wire harness. It consists of five individual modules which are combined into two so-called assemblies (Fig. 2). Contingent on engine configuration or country-specific adaptations, there are several variants of these modules and assemblies (certain combinations of these variants, however, do not occur).

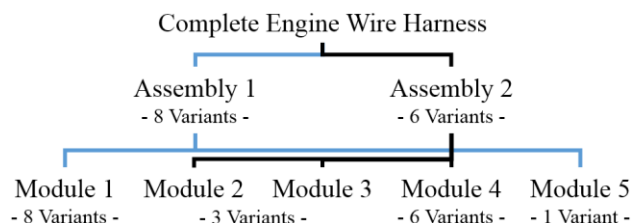


Fig. 2. OM654 Engine Wire Harness Variant Tree.

Fig. 1 also illustrates the variant diversity of selected components like plugs, fixations, wire channels, and sheaths.

The wire harness uses single wire seals or matt seals (Fig. 3) to guarantee media tightness. Twisted wires connect the motor control device with the injectors. Different tapes, fabric hoses and corrugated tubes bundle and protect the wires against abrasion, temperature as well as media. Cable channels and fixations ensure that the wire harness stays in place over the life span.

## 3. Wire Harness Design Process

The basic function of a wire harness is to ensure the reliable and error-free electrical connection between electronic components over its life span which is generally about 15 to 20 years. In order to achieve all required functions, the wire harness layout as well as the selection of suitable materials and

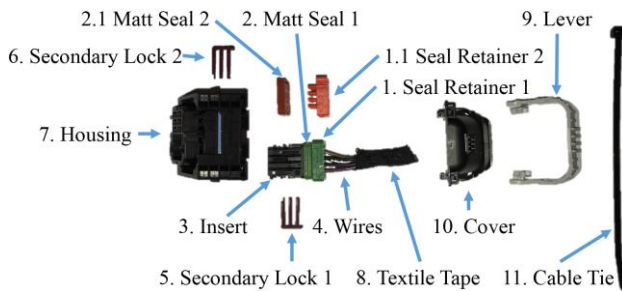


Fig. 3. Components of a Motor Control Device Plug.

components are of key importance. [8] In the following, the design process and some common design influences are described in more detail.

First, a wiring diagram is created, which lists all connections between electronic components, necessary voltages and currents as well as resulting wire diameters. In order to reduce electro-magnetic interference on electrical signals, sensitive control connections require utilization of twisted wire pairs instead of single ones [2]. Additionally, a 3D CAD layout is sculptured which takes not only component locations into account but also influencing factors like relative movements of components, installation spaces, sharp edges, temperature ranges, media resistance, installation requirements among others. These influencing factors usually lead to an adjusted layout and the usage of special-purpose (see Fig. 3), components. Based on the wiring diagram as well as the 3D CAD layout a detailed 2D engineering drawing is created. This 2D drawing is mapped onto the flat mounting board which is used to assemble the wire harness.

#### 4. Wire Harness Production Process

The wire harness production process chain can usually be divided into six stages shown in Fig. 4.

In preproduction basic process steps like wire cutting, stripping, mounting of single wire seals, terminal attachment and wire twisting take place. Manufacturing submodules such as plug preassembly, wire insertion and bundling are also common processes in preproduction.

All components or submodules are stored afterwards in intermediate buffers or warehouses. As soon as the parts are requested at the assembly line a delivery to the supermarket is triggered. [2, 3] The supermarket is a storage area close to the assembly line where all necessary parts like processed wires, plugs, submodules, attachments, etc. are stored in reach of the assembly workers.

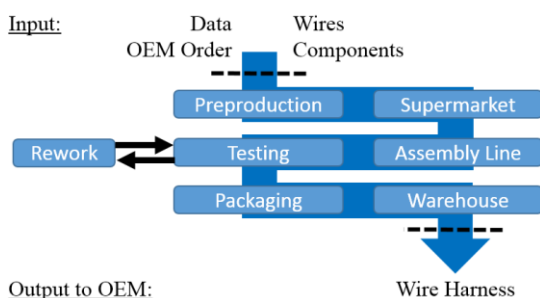


Fig. 4. Automotive Wire Harness Production Process Chain.



Fig. 5. Mounting Board in the Wire Harness Production.

At the assembly line, the most important and labor-intensive part of the wire harness manufacturing process, usually mounting boards (see Fig. 5), which circulate, are used. Assembly processes include, among others, completing the plugs, inserting wires in the connector housings, routing the wires on the mounting board, bundling the wires with corrugated tubes or wrapping them with adhesive tape. Finally parts like wire channels, cable ties and fixations are installed. [8]

After the assembly process, a 100% function and completeness test of the entire wire harness is mandatory. For this, the wire harness is taken off the mounting board and placed on a specially designed testing table. Since connectors are not allowed to be plugged in during this test, the testing table is equipped with special brackets which touch the terminals in the connector housings without using the plug mechanism. [8] Different tests are performed: These include an electric function test, a leakage test via air pressurization, a completeness check and in case plugs with secondary locks (Fig. 3) are used, a tight fit test of the locks [13]. In case of an error, a documented rework process has to be performed. For example an exchange of a single wire often results in removal of a large amount of tape or other attachments. For a replacement of a crimped terminal of the plug, special tools are available for opening the securing lock. After the rework process, another 100% function and completeness test of the reworked wire harness is mandatory.

After a successful tests, a label is printed and applied on the wire harness. Subsequent packaging and storage in the warehouse follows. [8]

#### 5. Automation in the Production Process

Wire harness assembly involves a high number of manual work steps. The degree of automation varies substantially between preproduction and assembly line. In preproduction, process steps like wire cutting, stripping, mounting of single wire seals, terminal attachment and wire twisting are well automated [8]. Selected special-purpose machines and their functions are listed in Table 1. In contrast, almost all work steps at the assembly line are executed manually. Typical work steps include wire routing, terminal insertion, bundling with e.g. cable ties, adhesive tape, corrugated tubes or plastic fabric hoses and attaching cable channels. Handheld appliances are available for bundling with standard cable ties and adhesive tape. The testing procedures on the testing table are mostly automated. However, wire harness handling, i.e. placing the wire harness on the testing table, untangling and fitting

Table 1. Wire Processing Machines and their Functions.

|                                | Cutting | Stripping | Seal | Insertion | Terminal Attachment | Wire Twisting | Single Wire Insertion | Twisted Wire Insertion |
|--------------------------------|---------|-----------|------|-----------|---------------------|---------------|-----------------------|------------------------|
| Komax „Kappa“                  | x       | x         |      |           |                     |               |                       |                        |
| Schleuniger „Strip-Series“     | x       | x         |      |           |                     |               |                       |                        |
| Schleuniger „UniCrimp“         |         |           |      |           | x                   |               |                       |                        |
| Schleuniger „StripCrimp“       |         | x         |      |           | x                   |               |                       |                        |
| Komax „Alpha“ / „Gamma“        | x       | x         | x    |           | x                   |               |                       |                        |
| Schleuniger „CrimpCenter“      | x       | x         | x    |           | x                   |               |                       |                        |
| Komax „bt“                     |         |           |      |           |                     | x             |                       |                        |
| Schleuniger „WireTwister 3300“ |         |           |      |           |                     | x             |                       |                        |
| Komax „Alpha 488 S“            | x       | x         | x    |           | x                   | x             |                       |                        |
| Komax „Zeta“ / „Omega“         | x       | x         | x    |           | x                   |               | x                     |                        |

connectors as well as attachment parts and fixtures in testing brackets, is done by hand.

## 6. Automation Requirements and Current Challenges

The most time-consuming processes in wire harness production are wire preparation (up to 38% of standard time), wire routing (up to 50%), and wire bundling (up to 30%) as studies from the 1980's and 1990's indicate [14, 15]. Unfortunately, no current data is publicly available [16]. Since most automation solutions operate on the preproduction level (Section 5), we expect the work shares in particular in the assembly line to coincide with these estimates.

The main obstacles for automation are the high variety of variants, the flexibility of limp work-pieces such as wires, the complexity and high number of parts within a wire harness, the lack of technical solutions, high investment costs and low flexibility of available special-purpose appliances [14, 15]. Despite of overall high production volumes, up to now one of the fundamental challenges remains the diversity of variants which in extreme cases results in customer-specific wire harnesses of batch size 1 [10, 17].

In general, a highly automated wire harness production regarding costs, quality, documented processes and traceability is preferred. This includes, for example, banning of manual crimping or splicing processes. The wire harness design severely influences the achievable degree of automation in its production (design to manufacture). Thus, partition of assemblies into modules (Section 2) should be such that wire insertion of most modules is fully automated.

In addition to the severe influence of the wire harness design on the production, available appliances have their own challenges. The challenge with these often rigid automatized appliances mainly used in preproduction is that, due to their restricted flexibility, minor changes like a simple color change of the wire, can lead to complex manual set-up processes. These include, among others, controlling the stripped length, crimp force, crimp height as well as executing a pull out test and sometimes a micrograph. In order to reduce costs, a low percentage of time for set-up is desired. This, in turn, leads to the build-up of high stocks.

Also highly automated appliances like a Komax Zeta 656 or Komax Omega (Table 1) are not able to accomplish their tasks if the variant diversity exceeds the limitations of the appliances, for example the number of different terminal types, seals, wire colors and diameters. An extension of the appliance with additional stations for a minimal increase of the degree of automation is usually rejected because of cost reduction.

There are no technical solutions available to insert twisted wire pairs into connector housings. To illustrate this circumstance, we compare Module 5 (see Fig. 1) of our exemplary wire harness with an injector module of a competitor (referred to as Module X). Module 5 consists of twisted wires, different tapes, plastic fabric hoses, cable ties, five plugs and a cable channel. To increase the flexibility in the area near to the four injection plugs plastic fabric hoses are used. Because there is no technical solution available to insert twisted wire pairs and the fabric hoses have to be pulled over the wires prior to wire insertion, the corresponding wires of Module 5 have to be inserted by hand. A replacement of the hoses with tape would cause a slight deterioration of flexibility but reduce the number of parts and working steps. In contrast, Module X has no twisted wires and uses slotted corrugated tubes instead of tapes. Therefore, a Komax Zeta 656 or Komax Omega can be used for wire preparation and insertion. If the corrugated tubes and other attachments are mounted directly in the preproduction it is even conceivable that Module X never has to be put on a mounting board.

With regard to the assembly line, there is no equipment available on the market which can fully automate the various manual work steps. Additionally, analyzing the value stream of wire harness manufacturing process indicates that there is a lot of waste and also many necessary non-value-adding processes. Aiming at a fully automated manufacturing of wire harnesses the entire production process has to be taken into consideration. A simple replacement of manual work steps without regarding the surrounding conditions will not lead to an optimal solution because non-value adding processes or waste further exist.

Today's production processes still have to struggle with all previously listed automation restraints like high variety of variants, limp behavior of the wires, high number of parts within a wire harness, lack of technical solutions, high investment costs and low flexibility of the appliances [14, 15]. Also the wire harness design has a strong influence on the production process (design to manufacture). There is no single automation restraint; instead there is a conglomerate of competing restraints. A suitable automation solution, therefore, requires a holistic view of the entire production process.

## 7. Innovations and Current Research

As early as the 1980's, automation solutions for issues like configuration of the mounting board, positioning of connector housings and wire routing were investigated at the Fraunhofer IPA in Stuttgart [14, 18]. Further, the preparation and insertion of wires with crimped terminals in connector housings using a robot was prototyped [19]. Automation of wire routing remains as a field of research [20]. At the Institute for Factory Automation and Production Systems (FAPS) in Erlangen, a flexible robot, programmed by manual guided movement with

a special, partly rapid prototyped, gripping system to grasp, route and insert wires, is used [21].

Fully automated wire bundling with adhesive tape was investigated at IPA in the 1990's and still remains as an area of research [15, 22]. In 2018, a ready-for-use spot taping robot by KABATEC was presented.

A procedure and appliance correcting the position of the free wire ends of twisted wire pairs is developed [23]. Knowledge of the correct wire end position is crucial for the automated insertion in connector housings. Another way is to twist the wires after the insertion process with an automatic placement machine addable appliance [24].

Installing a wire harness in an instrument panel with a three arm robot including process steps like picking up a hanging wire harness and inserting eight connectors was investigated. Visual markers were used for connector recognition and localization. [25]

To control the correctness of assembly processes an image processing system can be used [26]. It is also conceivable to control the presence of attachments by implementing an automatic inspection system [27].

A concept for a complete wire harness production includes the production steps wire cutting, stripping and crimping realized by one crimping machine as well as placement of connector housings on the mounting board, wire routing, contact insertion, wrapping and harness removal executed by four different robots [28].

More fundamentally, the substitution of an instrument panel wire harness with a large-area flexible printed circuit (FPC) was investigated. But a FPC has to face challenges like the dependence between interconnection task and physical architecture. Therefore, the authors conclude that a direct replacement seems not feasible and at first the development of FPCs to substitute smaller and less complex automotive wire harnesses is necessary. [29]

## 8. Outlook and Necessary Research

With a fully automated wire harness production in mind, the development of appliances that automate the various manual

work steps at the assembly line has great potential. Table 2 lists typical manual work steps at the assembly line, past and current research as well as pilot stations. Furthermore, adjacent fields of research that may support the wire harness related research but may severely differ regarding above mentioned requirements like relative movements of components, installation spaces, sharp edges, temperature ranges, media resistance as well as installation requirements among others.

This overview highlights wire routing, wire insertion, bundling with cable tie as well as adhesive tape bundling as important necessary fields of further research. It is expected, that focused research in these fields will lead to substantial replacement of manual work steps with automated processes.

## 9. Future Requirements

'Connected', 'Autonomous', 'Shared & Services' and 'Electric' (CASE) are keywords for future automotive trends seen by Daimler AG [40]. Especially autonomous driving, electrification as well as additional trends like miniaturization, higher data rates and high-voltage harnesses will have a great influence on the production process of automotive wire harnesses [9, 11]. For autonomous driving, the wire harness will become a safety relevant part which introduces stricter requirements on reliability and redundancy of electrical connections. Due to the increase in functionality and weight reduction wire diameters and terminal sizes will become smaller. [6] This miniaturization will require a fully automated insertion and rework process because of risk of damage. All these trends will require more documentation, traceability and monitoring of the production process. These requirements can only be met with a higher degree of automation [10].

For German car manufacturers Romania or Poland are still attractive wire harness production locations, due to transportation times by truck of less than one day. Also, the state fragility categorization is "more stable" instead of "elevated warning" like for Tunisia and Morocco [41]. Increasing real wages between 4% and over 5% per year in Romania and Poland promote development as well as usage of automation to keep costs on a stable level [6, 42].

Table 2. Manual Assembly Line Work Steps and Appendant Research.

| Task               | Component               | Research     | Pilot Station                                                   | Adjacent Research Field                                                                           |
|--------------------|-------------------------|--------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Routing            | Wire                    | [14, 18, 21] | [14, 18, 21]                                                    | Rittal - Averex Verdrahtungszentrum<br>System Robot Automazione – Syndy; [30, 31]                 |
| Insertion          | Wire                    | [19, 21]     | [19, 21]                                                        | [32, 33]                                                                                          |
| Placing            | Connector Housing       | [14]         | [14]                                                            | IPA – Separation Processes Using Robots: Bin Picking                                              |
| Placing            | Preamsembled Harness    | -            | -                                                               | [25, 34–37]                                                                                       |
| Assembly           | Plug                    | -            | -                                                               | Axmann Technology                                                                                 |
| Bundling           | Cable Tie               | [14]         | Shenzhen Swift Automation Technology - Robot with Cable Tie Gun | -                                                                                                 |
| Bundling           | Adhesive Tape           | [15]         | Komax - Spot Taping Robot; [15]                                 | [38]                                                                                              |
| Bundling           | Tube, Hose              | -            | -                                                               | [39]                                                                                              |
| Attaching          | Fixation, Cable Channel | -            | -                                                               | RNA Automation – SIX Axis Robot Clipping & Welding System                                         |
| Completeness Check | Wire Harness            | -            | -                                                               | IPA – Motion Analysis for Quality Assurance<br>TrueSolttec - USB Camera Cable Color Checker T-CAM |



## 10. Conclusion

A continuous automation solution has to face many challenges illustrated by a supposedly simple example of a wire harness. These include mainly the high diversity of variants, the flexibility of limp work-pieces, the complexity and high number of parts within a wire harness, the lack of technical solutions, high investment costs and low flexibility of available special-purpose appliances [14, 15]. Automation solutions already developed in the 1980's never passed pilot status as current production process depicts. Today, almost all work steps at the assembly line are executed manually, while there are many specialized appliances in preproduction. Latest trends and resulting requirements can only be met by an increasing degree of automation in wire harness production.

There is a complex bundle of restraints for the automation of the wire harness production, but two major ways can be defined. On the one hand, the development of additional solutions for subareas. On the other hand, a completely new approach for the production of wire harnesses may be a chance to solve the problem.

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